

MECHANISM OF SPARK PLASMA SINTERING

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Abstract

Spark plasma sintering (SPS) is a pressure sintering method based on high temperature plasma (spark plasma) momentarily generated in the gaps between powder materials by electrical discharge at the beginning of ON-OFF DC pulse energizing. The large current pulse energizing method generates: (1) spark plasma, (2) spark impact pressure, (3) Joule heating, and (4) an electrical field diffusion effect. This sintering mechanism and mechanical properties of SPS sintered compact show different characteristics compared to conventional pressure assisted sintering processes. The process offers significant advantages with various kinds of new materials and consistently produces a highly dense compact in a shorter sintering time and of finer grain than conventional methods.

This paper introduces SPS systems, principles of processing, features and examples of applications.

1. Introduction

Spark plasma sintering (SPS) is a newly developed process—a synthesis and processing technique—which makes possible sintering and sinter-bonding at low temperatures and short periods by charging the intervals between powder particles with electrical energy and effectively applying a high temperature spark plasma generated momentarily. It is regarded as a rapid sintering method, using self-heating action from inside the powder, similar to self-propagating high temperature synthesis (SHS) and microwave sintering. SPS systems offer many advantages over conventional systems using hot press (HP) sintering, hot isostatic pressing (HIP) or atmospheric furnaces, including ease of operation and accurate control of sintering energy as well as high sintering speed, high reproducibility, safety and reliability. The SPS process is expected to find increased use in the fabrication of functionally graded materials (FGMs), intermetallic compounds, fiber reinforced ceramics (FRC), metal matrix composites (MMC) and nanocrystalline materials, which are difficult to sinter by conventional sintering methods. Figure 1 shows the materials covered by SPS processing.

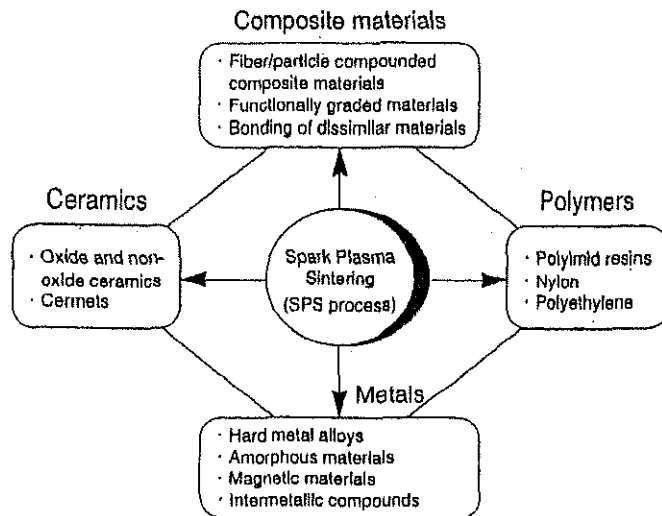


Figure 1 Typical example of materials covered by SPS processing
(Source: SPS Forum survey)

The history of the technology related to the process in question started in the late 1930s when a sintering process using electrical energizing was introduced in the United States. In Japan, a similar process based on the pulse current applied sintering method was researched and patented in the 60s and is known as spark sintering [1] [2] , but it was not put to wide use due to the lack of application technology at that time, limited fields where it could be applied and unsolved problems associated with industrial production, equipment cost and sintering efficiency. There was little literature on research into this process until the latter half of the 70s. The second generation was developed from the middle of the 80s to the early 90s. These units were small experimental systems—Plasma Activated Sintering (PAS) with maximum sintering pressure of around 5 tons and pulse generators of up to 800 amp, used primarily for materials research. However, after the recent debut of the Spark Plasma Sintering (SPS) process [3] as the third generation of this advanced technology, the SPS system with large DC pulse generators of 10 to 100 tons and 2,000 to 20,000 amp and more (Fig. 2) gained a reputation as new industrial processes for synthetic processing of gradient and composite materials [4] [5]. The process recently has attracted growing attention among production engineers as well as material researchers.

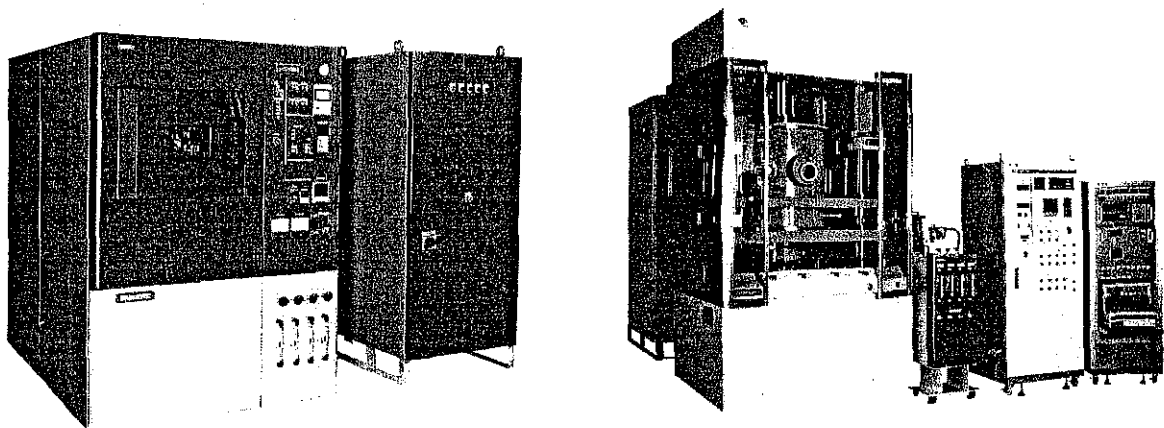


Figure 2 10-ton medium (left), 100-ton large-sized (right) spark plasma sintering systems

2. Principles of the SPS process

The SPS process features a very high thermal efficiency because of the direct heating of the sintering graphite mold and stacked powder materials by the large spark pulse current. It can easily consolidate a homogeneous, high-quality sintered compact because of the uniform heating, surface purification and activation made possible by dispersing the spark points.

2.1 Basic configuration of the SPS system

Figure 3 shows the basic configuration of a typical SPS system.

The system consists of a SPS sintering machine with a vertical single-axis pressurization mechanism, specially designed punch electrodes incorporating water cooler, a water-cooled vacuum chamber, a vacuum/air/argon-gas atmosphere control mechanism, a special DC-pulse sintering power generator, a cooling water control unit, a position measuring unit, a temperature measuring unit, an applied pressure display unit and various interlock safety units. Table 1 shows an example of suitable materials for SPS processing. Figure 4 shows the inside of the water cooling chamber during spark plasma sintering.

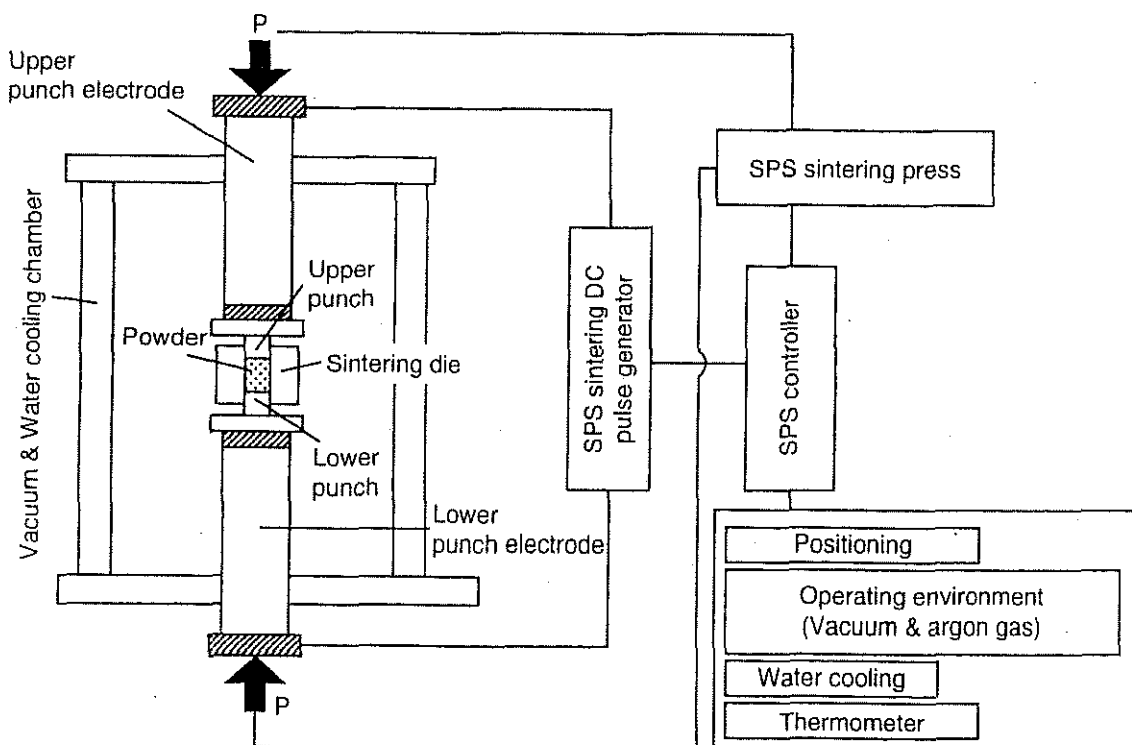


Figure 3 SPS system configuration

Table 1 Suitable materials for SPS processing

Classification		Materials for SPS processing
Metals		Fe, Cu, Al, Au, Ag, Ni Cr, Mo, Sn, Ti, W, Be Virtually any metal possible
C e r a m i c s	Oxides	Al_2O_3 , Mullite, ZrO_2 , MgO, SiO_2 , TiO_2 , HfO_2
	Carbides	SiC, B_4C , TaC, TiC, WC, ZrC, VC
	Nitrides	Si_3N_4 , TaN, TiN, AlN, ZrN, VN
	Borides	TiB_2 , HfB_2 , LaB_6 , ZrB_2 , VB_2
	Flourides	LiF, CaF_2 , MgF_2
Cermets		$\text{Si}_3\text{N}_4 + \text{Ni}$, $\text{Al}_2\text{O}_3 + \text{Ni}$, $\text{ZrO}_2 + \text{Ni}$ $\text{Al}_2\text{O}_3 + \text{TiC}$, SUS + ZrO_2 , $\text{Al}_2\text{O}_3 + \text{SUS}$ SUS + WC/Co, BN + Fe, WC + Co + Fe
Intermetallic copounds		TaI, MoSi_2 , Si_3Zr_5 , NiAl NbCo, NbAl, LaBaCuSO_4 , $\text{Sm}_2\text{Co}_{17}$
Other materials		Organic materials (polyimide, etc.), composite materials

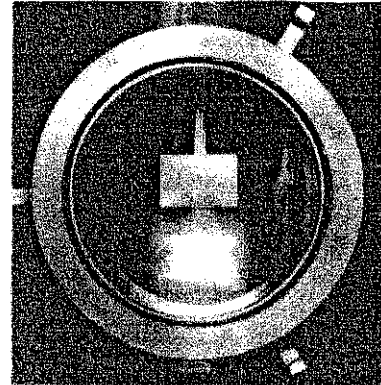


Figure 4

Sintering stage in SPS vacuum chamber

2.2 DC pulse current energizing effect

The ON-OFF DC pulse energizing method generates: (1) spark plasma, (2) spark impact pressure, (3) Joule heating, and (4) an electrical field diffusion effect. In the SPS process, the powder particle surfaces are more easily purified and activated than in conventional electrical sintering processes and material transfers at both the micro and macro levels are promoted, so a high-quality sintered compact is obtained at a lower temperature and in a shorter time than with conventional processes.

Figure 5 illustrates how pulse current flows through powder particles inside the SPS sintering die.

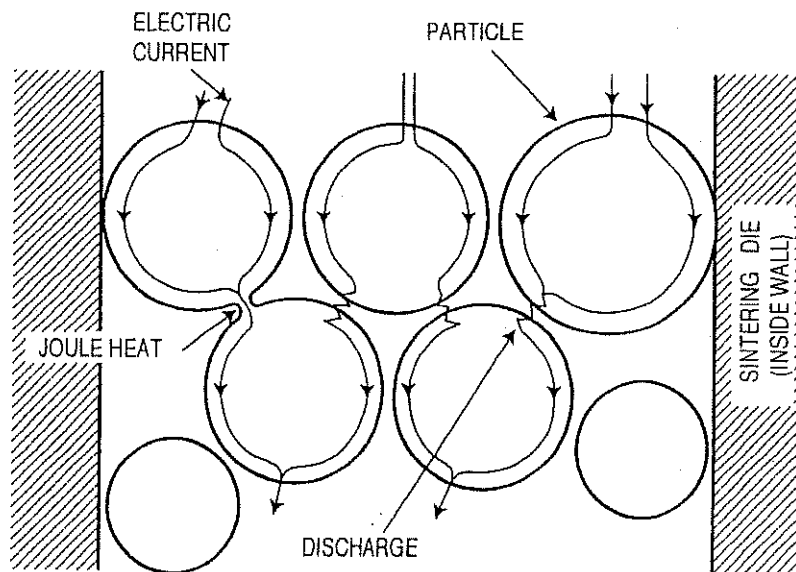


Figure 5 Pulsed current flow through powder particles

Conventional electrical hot press processes use DC or commercial AC power, and the main factors promoting sintering in these processes are the Joule heat generated by the power supply (I^2R) and the plastic flow of materials due to the application of pressure. The SPS process is an electrical sintering technique which applies an ON-OFF DC pulse voltage and current from a special pulse generator to a powder of particles, and in addition to the factors promoting sintering described above, also effectively discharges between particles of powder occurring at the initial stage of the pulse energizing for sintering. High temperature sputtering phenomenon generated by spark plasma and spark impact pressure eliminates adsorptive gas and impurities existing on the surface of the powder particles. The action of the electrical field causes high-speed diffusion due to the high-speed migration of ions. The application of the pulse voltage induces various phenomena as shown in Figure 6.

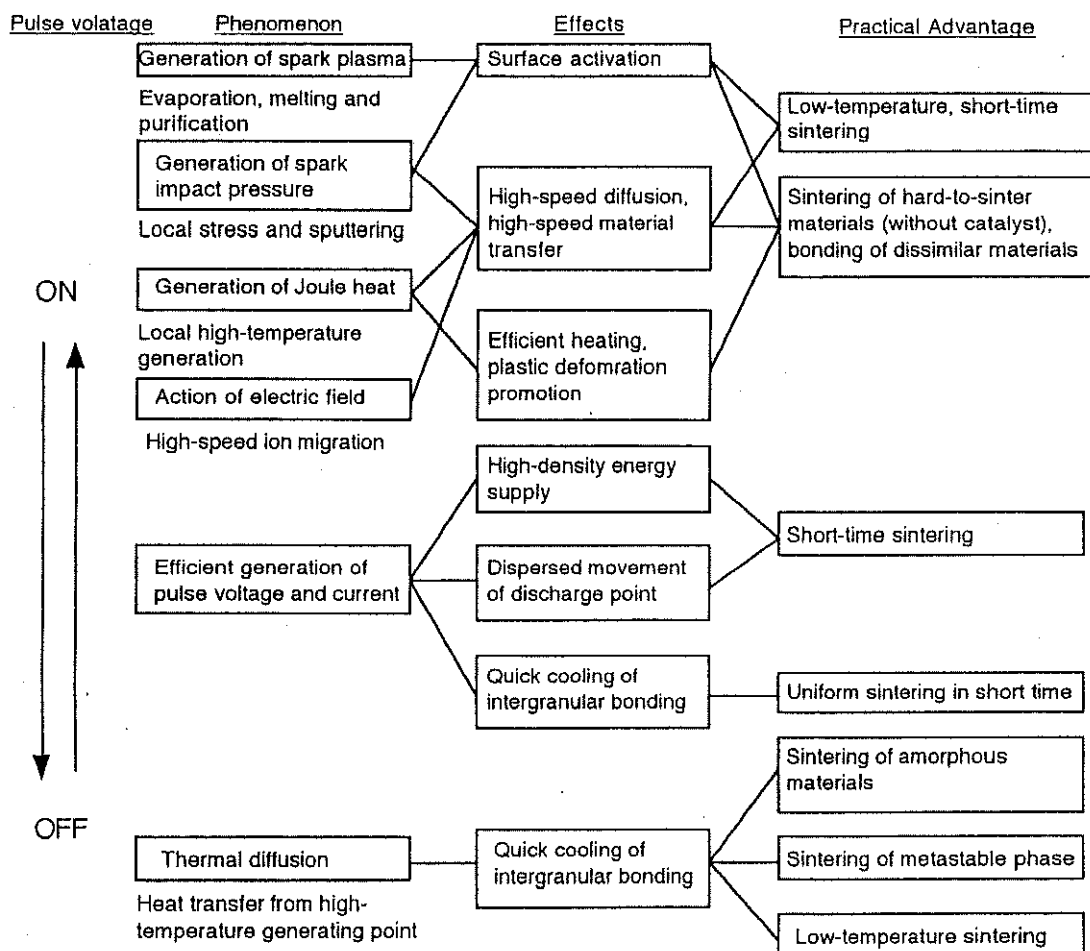


Figure 6 Effect of ON-OFF DC pulse energizing

2.3 Mechanism of processing

When a spark discharge appears in a gap or at the contact point between the particles of a material, a local high temperature-state (discharge column) of several to ten thousands of degrees centigrade is generated momentarily. This causes evaporation and melting on the surface of powder particles in the SPS process, and "necks" are formed around the area of contact between particles. Figure 7 shows basic mechanism of neck formation by spark plasma.

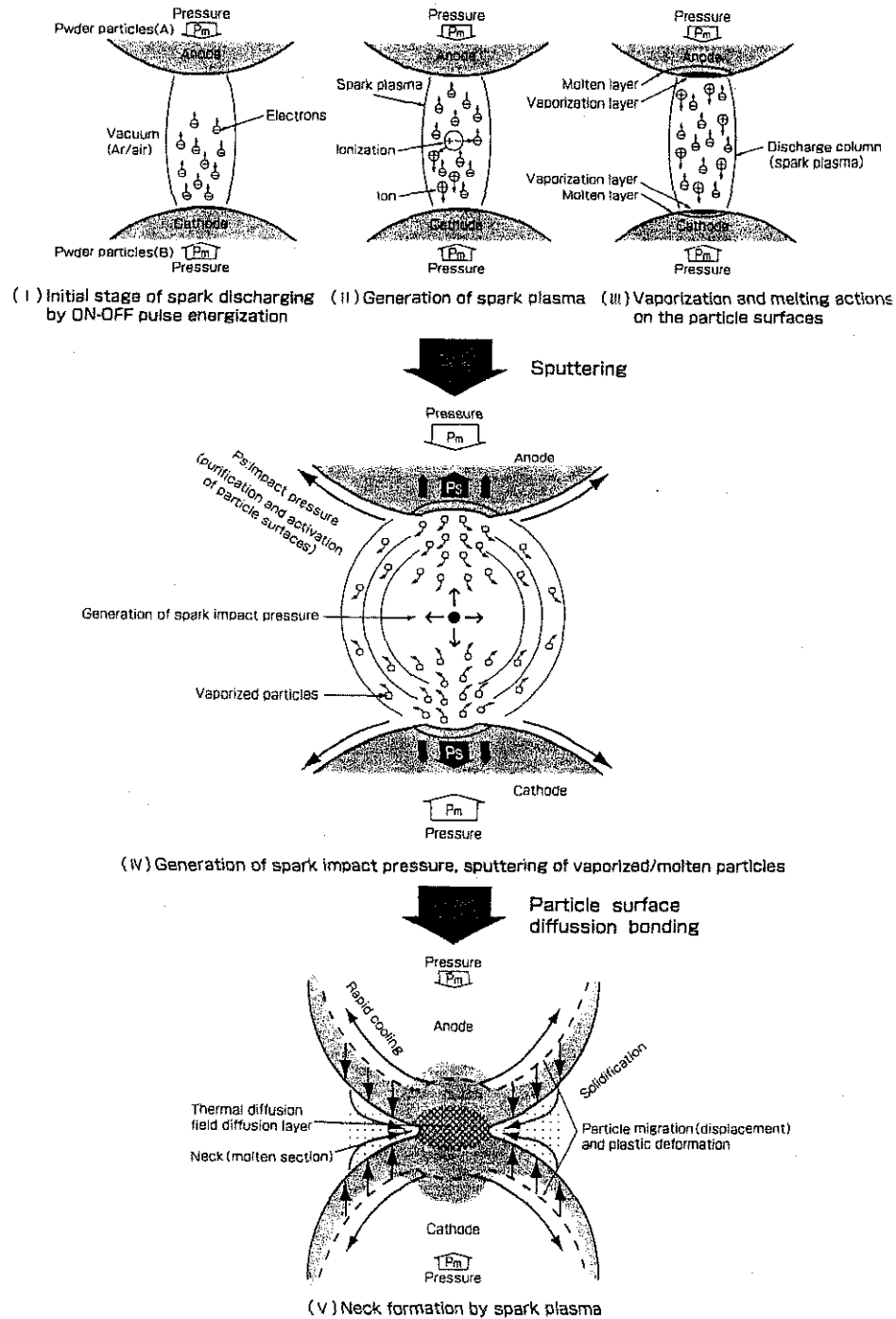


Figure 7 Basic mechanism of neck formation by spark plasma

Figures 8 and 9 show a typical ON-OFF DC pulsed current path by the SPS and a typical conventional material transfer path mechanism of evaporation, solidification, volume diffusion, surface diffusion and grain boundary diffusion on neck formation.

Figures 10, 11 and 12 are SEM micrographs showing the results of SPS experiments performed at normal atmospheric pressurizing a sintering die and punches made of a graphite and a spherical bronze alloy powder. Figure 10 shows the behavior in the initial stage of neck formation due to sparks in the plasma. The heat is transferred immediately from the center of the spark discharge column to the sphere surface and diffused so that intergranular bonding portion is quickly cooled. As seen in Figure 11 which show several necks, the pulse energizing method causes spark discharges one after another between particles. Even with a single particle, the number of positions where necks are formed between adjacent particles increases as the discharges are repeated.

Figure 12 shows the condition of an SPS sintered grain boundary which is plastic-deformed after the sintering has progressed further. This state is the result of the processing conditions in which the applied pressure was 29 MPa, the sintering temperature was 500°C, the holding time was 120 sec., the SPS current 850 amp, and the voltage was 3.9 v.

The sintering dies and punches made of graphite are subject to Joule heating according to the progress of the sintering of the internal powder material, and function as heating elements to assume the role of maintaining the homogeneousness of the sintering temperature for the densification.

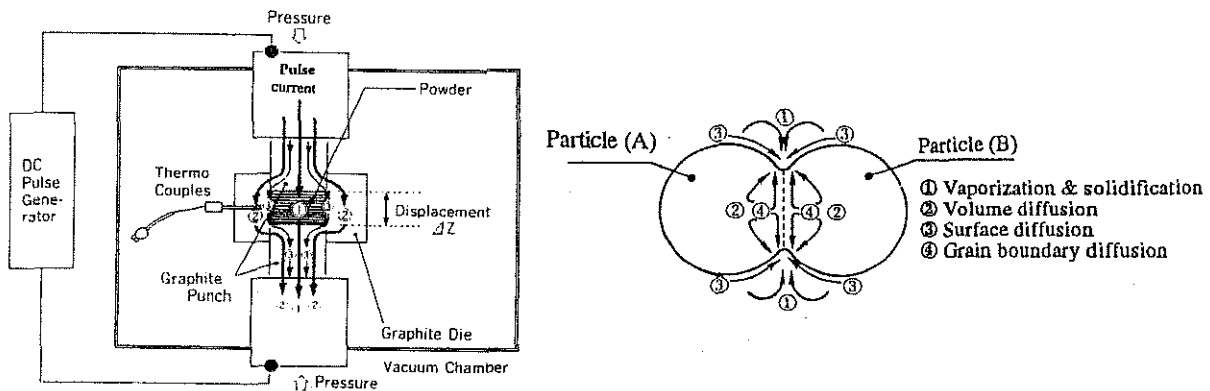


Figure 8 ON-OFF DC pulsed current path Figure 9 Material transfer path model during sintering



Figure 10 Initial stage of neck formation

Figure 11 Expansion of neck

Figure 12 Start of plastic flow

Figure 13 shows the influences of the applied pressure and sintering temperature of the SPS process on intergranular bonding. The photos show the SPS sintering states observed with an optical microscope in different samples; a sample in which the sintering temperature was constant at 600° C and the applied pressure was the parameter which was varied between 13, 29 and 49 MPa and a sample in which the applied pressure was constant at 29 MPa and the sintering temperature varied between 500, 600 and 700° C. The holding time was 120 seconds in both cases. The required sintered material can be provided with the optimum degree of bond basically by combining the plastic deformation force (applied pressure) and heat (temperature at the measuring point on the outer wall of the die). The intergranular bonding inside the powder becomes porous at lower pressures and lower temperatures, and the grain boundaries decrease at higher pressures and higher temperatures.

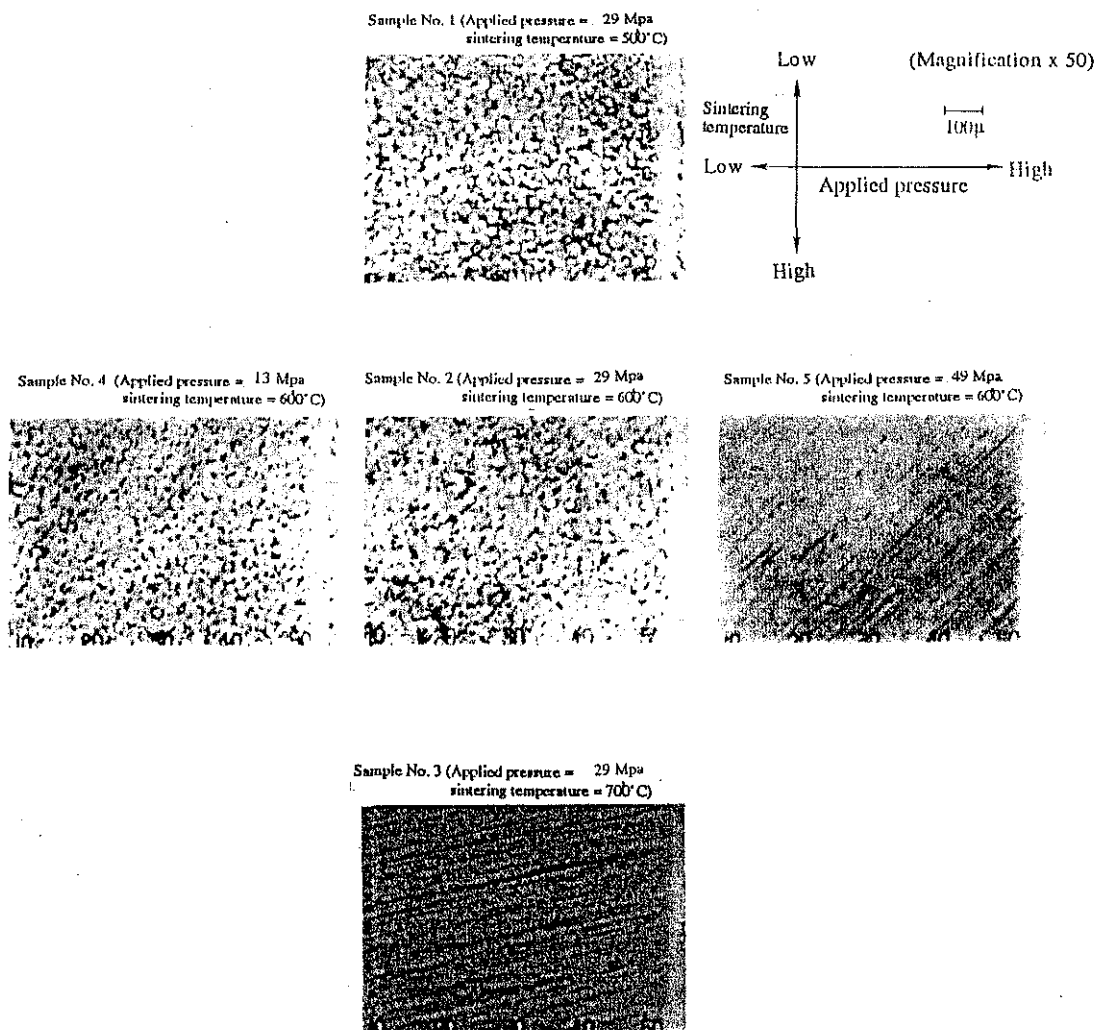


Figure 13 Influence of pressurizing and sintering temperature condition of SPS

3. Example of SPS process applications

High-temperature short-period SPS sintering is expected to provide almost all ceramic materials with new characteristics and sintered effects which are different from those obtained by the HP and HIP processes [6] [7] [8].

The ceramic materials which can be sintered at high density include oxides such as Al_2O_3 , mullite, ZrO_2 , MgO , HfO_2 and SiO_2 , carbides such as SiC , B_4C , TaC and TiC , borides such as TiB_2 and HfB_2 and nitrides such as Si_3N_4 , TaN , TiN and AlN .

3.1 Synthesis example of silicon nitride Si_3N_4

With the SPS process, a highly-dense sintered sample of silicon nitride ceramics can be fabricated at a much lower temperature and in a shorter holding time compared with hot pressing.

Figure 14 is a SEM micrograph of $\alpha\text{-Si}_3\text{N}_4$ starting powder material. It shows that the structure is composed of submicron isotropic particles (alpha phase, approx. 91%) and acicular particles beta phase, 9%) containing 5 weight % magnesia and 5 weight % of yttria as additives.

Figure 15 shows the micro structure of a SPS sintered specimen. With an applied pressure of 29 MPa, a sintering temperature of 1500 to 1600° C and a processing time totalling 12 minutes, composed of a 5-minute heating up time and 7-minute holding time, a sintered compact with a relative density of 98.7% and hardness of 18.3 GPa was obtained. The results of x-ray diffraction in Figure 16 shows that the beta phase has increased by approx. 6% by weight from the starting material and grain growth was minimized.

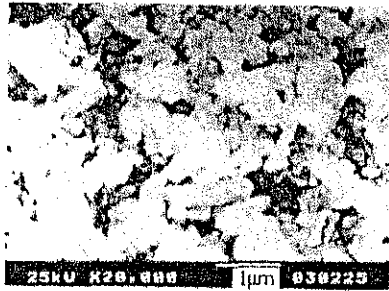


Figure 14 Si_3N_4 starting powder material

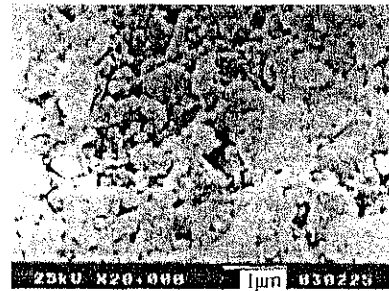


Figure 15 Microstructure of SPS sintered compact (Polished-etched surface)

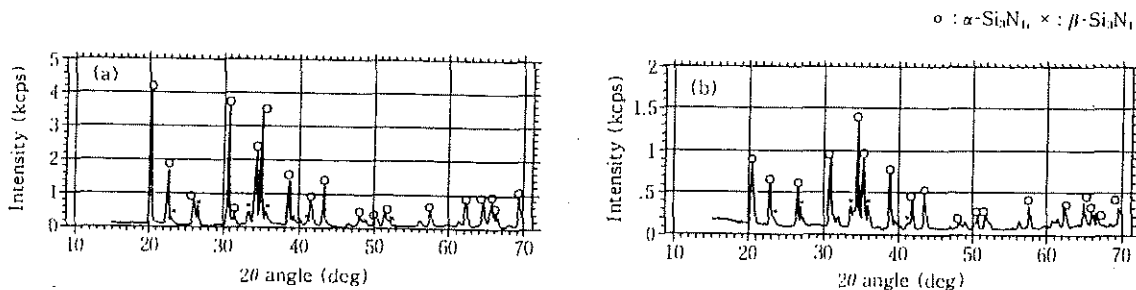


Figure 16 X-ray diffraction patterns:

(a) Starting powder material

(b) Sintered compact

3.2 Example of silicon carbide (SiC)

Figure 17 shows an example of sintered compact of ultrafine silicon carbide material (SiC). The starting powder materials is 99% purity without sintering additives. It can be consolidated to more than 99% of the theoretical density using SPS ultrahigh-temperature sintering. The SPS heating up and holding time totalled only 7 minutes. The density of this material which can be achieved by conventional processes has been up to 92 to 93% of the theoretical density. The sintered compact by spark plasma sintering has better mechanical properties than conventional sintered materials, a micro-Vickers hardness of 28.6 GPa and a fracture toughness of $4.7 \text{ MPa}\cdot\text{m}^{0.5}$ as shown in Table 2. The result of a comparison of x-ray diffraction patterns show that the properties of the starting material are maintained in the sintered compact (Figure 18). No pores or granular growth were observed and inspection of the fracture surface with SEM shows transgranular fractures which indicates strong intergranular bonding.

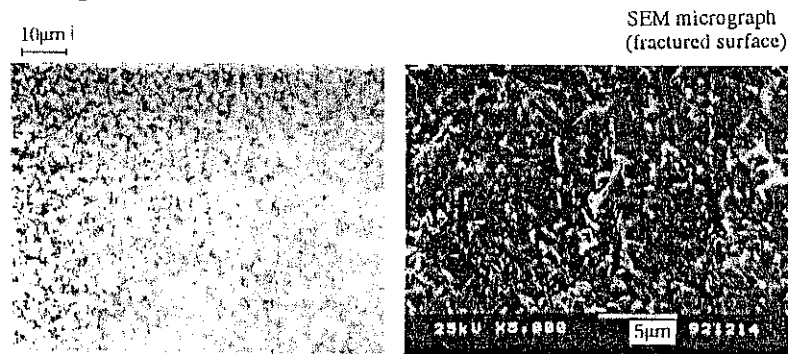
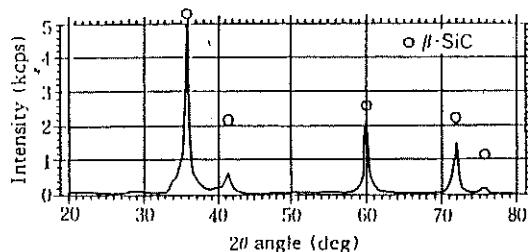


Figure 17 Optical micrograph of the etched surface (left) and SEM micrograph (right) of an ultrafine pure SiC sintered material

Table 2 Comparison of mechanical properties of an ultrafine pure SiC sintered compact.

Measured Item	SPS Sintered Compact	Conventional Compact
Micro-Vickers hardness (GPa)	28.6	23.0 ~ 29.0
Fracture toughness ($\text{MPa}\cdot\text{m}^{0.5}$)	4.7	3.2 ~ 4.2

Starting powder material of ultrafine pure SiC powder



Sintered compact of ultrafine pure SiC

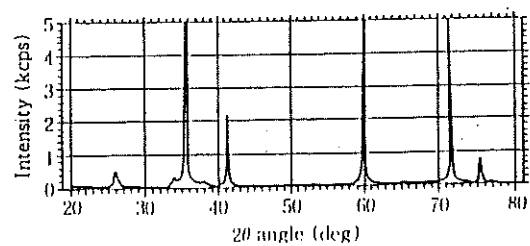


Figure 18 X-ray diffraction patterns: Starting powder material (left) Sintered compact (right)

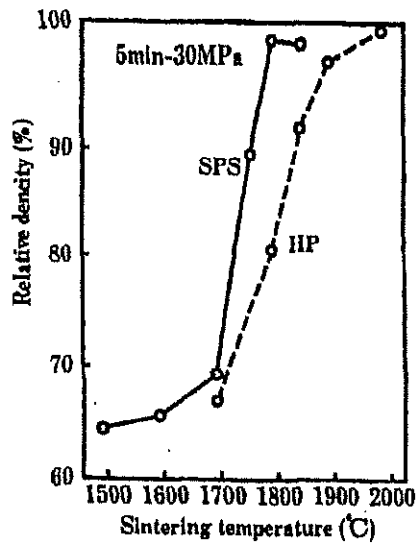


Figure 19 Densification behavior of silicon carbide by spark plasma sintering (SPS) and hot-pressing (HP)

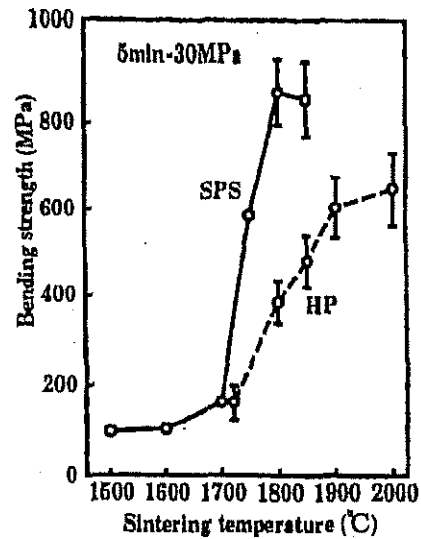


Figure 20 Effect of sintering temperature on bending strength at room temperature

Figures 19 and 20 show examples of silicon carbide compact prepared under the sintering condition of 30 Mpa and 5 minutes by the spark plasma sintering (SPS) method. The average starting powder particle size is $0.28\mu\text{m}$ containing 5 weight % of Al_2O_3 and 2 weight % of Y_2O_3 . Mechanical properties at room temperature were examined. The SPS fabricated dense silicon carbide ceramics at a sintering temperature of 1800°C lower than that of the hot-pressing process. The silicon carbide obtained by SPS had higher strength and fracture toughness than those obtained by hot-pressing[6].

3.3 Preparation of stainless steel/ ZrO_2 (3Y) Functionally Graded Material

Figure 21 shows the cross-sectional micrograph of the bonded section of a ZrO_2 stainless steel graded material fabricated by the SPS temperature-gradient field sintering process. The metal layer is composed of a stainless steel powder with an average particle size of $3\mu\text{m}$, and the ceramic layer is composed of submicron zirconia powder to which 3% by weight of yttria is added. By stacking three kinds of mixed-composition powders with stainless steel/zirconia ratios of 3/1, 1/1 and 1/3 between the 100% front and back layers, a total of 5 gradient layers were formed. Then DC pulse voltages from 5 to 1 v were applied, it was held at 1250° for 5 minutes, the pulse energizing was stopped and the specimen was cooled.

Observing the disk shaped sintered compact with a diameter of 20 mm and thickness of 3.2 mm obtained by this process, no pores or cracks were detected, and the hardness of the zirconia side surface was 13.5 GPa, indicating strong sinter-bonding. This fabrication proved that short-time sintering using a temperature gradient field at low-temperature can noticeably reduce the generation of residual stress in gradient materials.

We also succeeded in fabricating a ZrO_2/TiAl , ZrO_2/Ni , $\text{Al}_2\text{O}_3/\text{Ti}$, $\text{Al}/\text{Polyimide}$ and $\text{Cu}/\text{Polyimide}$ functionally graded material by the same method. Figure 22 shows typical examples of bulk FGM compacts fabricated by spark plasma sintering.

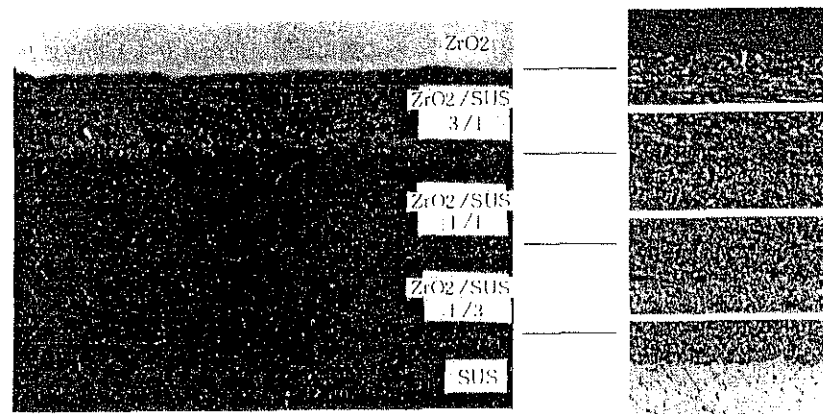


Figure 21 SPS sintered specimen with gradient bonding of zirconium oxide ZrO_2 (3Y) (upper) and stainless steel (lower). The photos on the right shows a magnified cross-section of each interface.

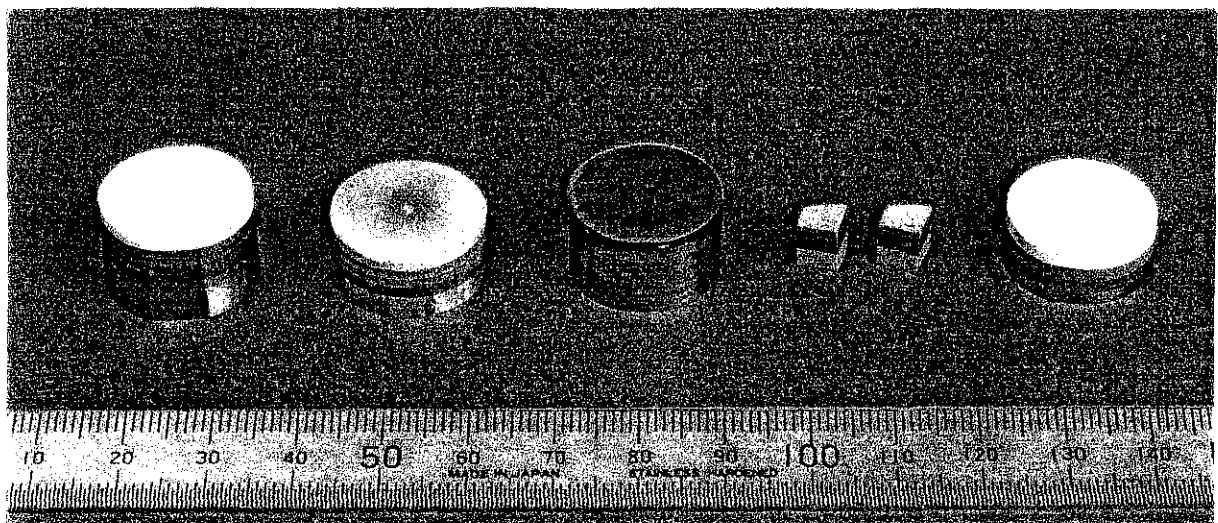


Figure 22 Typical examples of bulk FGM compacts fabricated by SPS.

From left: $\text{ZrO}_2(3\text{Y})/\text{stainless steel}$ with 6 interlayers,

$\text{ZrO}_2(3\text{Y})/\text{nickel}$ with 7 interlayers,

copper/stainless steel with 5 interlayers,

aluminum/polyimide with 3 interlayers and

$\text{Al}_2\text{O}_3/\text{titanium}$ with 3 interlayers.

4. Conclusion

In the above article, we have introduced the mechanism of SPS and some examples of ceramics and functionally graded materials applications. As a result of demand in an age of new materials which began in the 80s and the systematization of software and hardware over recent years, we are for the first time in the 90s, seeing that the technical value of the SPS process has finally started to be appreciated as a practical means for experimental use and industrial level production. SPS applications for functionally graded materials can be regarded as one noticeable example of this trend.

There are still a variety of R&D issues to be solved before the full potential of the SPS process is realized. These issues include systems automation, increasing the pulse current capacity and versatility of functions in the hardware, and the creation of a sintering technology data base and improvements to reliability and reproducibility in terms of software. In addition, for the SPS process to achieve the position of an integral synthetic processing technique for factory use, it is extremely important to develop suitable powder materials and establish fabrication techniques according to the SPS applications so that they match the characteristics of the new sintering process and systems.

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